LAB EXPERIMENT 1

M Vaishnavi 192324162

Experiment 1: Regular Expressions in Python

Aim

To write a Python program using Regular Expressions to match and search for patterns in text.

Algorithm

1. Start the program.
2. Import the re module.
3. Define a text string containing words and numbers.
4. Use re.search() to search for a pattern in the text.
5. Use re.match() to check a pattern at the beginning of the text.
6. Use re.findall() to find all occurrences of a pattern.
7. Use re.I for case-insensitive searching.
8. Search for digits using the regular expression \d+.
9. Display the results and stop the program.

Code

```
import re

text = "Python is easy. I am learning Python and Java."

# 1. Search for a pattern
result = re.search("Python", text)
print("Search:", result.group() if result else "Not found")

# 2. Match at the beginning
result = re.match("Python", text)
print("Match:", result.group() if result else "Not found")

# 3. Find all occurrences
result = re.findall("Python", text)
print("Find All:", result)

# 4. Case-insensitive search
result = re.findall("python", text, re.I)
print("Ignore Case:", result)

# 5. Search for digits
text2 = "My age is 20"
result = re.search(r"\d+", text2)
print("Digits:", result.group() if result else "Not found")
```

Output / Screenshot

```
[1]
✓ 0s
```

```python
import re

text = "My name is Ravi. My phone number is 9876543210."

# 1. Match a pattern at the beginning of the text
result = re.match("My", text)
if result:
    print("Match found:", result.group())
else:
    print("No match found")

# 2. Search for a pattern anywhere in the text
result = re.search("Ravi", text)
if result:
    print("Search found:", result.group())
else:
    print("Pattern not found")

# 3. Find all numbers in the text
result = re.findall(r"\d+", text)
print("Numbers found:", result)
```

```
Match found: My
Search found: Ravi
Numbers found: ['9876543210']
```

Result

Thus, the Python program to demonstrate Regular Expressions using re.search(), re.match(), and re.findall() was executed successfully, and the required patterns were matched and searched in the text.